

Clinical Reference: Complete Blood Count & Chemistry Panels

1. Complete Blood Count (CBC) Reference Intervals

Hemoglobin (Hgb):

- Adult Males: 13.8 to 17.2 g/dL
- Adult Females: 12.1 to 15.1 g/dL
- Low values indicate anemia; elevated values may suggest polycythemia or dehydration.

White Blood Cell Count (WBC): 4,500 to 11,000 cells/mcL.

- Elevated (Leukocytosis): Infection, inflammatory state, or physiological stress.
- Decreased (Leukopenia): Bone marrow suppression, viral illness, or autoimmune conditions.

Platelet Count: 150,000 to 450,000 /mcL (Thrombocytopenia vs Thrombocytosis).

2. Comprehensive Metabolic Panel (CMP) & Renal Function

Serum Creatinine: 0.7 to 1.3 mg/dL (adult males), 0.5 to 1.1 mg/dL (adult females).

eGFR (estimated Glomerular Filtration Rate): ≥ 90 mL/min/1.73m² (Normal).

Blood Urea Nitrogen (BUN): 7 to 20 mg/dL.

Electrolytes: Sodium (135-145 mEq/L), Potassium (3.5-5.0 mEq/L), Chloride (96-106 mEq/L).

3. Lipid Profile & Cardiovascular Risk Ratios

Total Cholesterol: Desirable below 200 mg/dL; Borderline high 200-239 mg/dL; High ≥ 240 mg/dL.

LDL Cholesterol ('bad' cholesterol): Optimal < 100 mg/dL; Borderline 130-159 mg/dL; High 160-189 mg/dL.

HDL Cholesterol ('good' protective cholesterol): Desirable ≥ 40 mg/dL (males), ≥ 50 mg/dL (females).

Triglycerides: Normal < 150 mg/dL; Borderline high 150-199 mg/dL; High ≥ 200 mg/dL.

4. Patient Guidance on Laboratory Deviations

A value slightly outside reference intervals does not automatically indicate illness.

Physicians evaluate trends over time alongside physical exams and clinical symptoms.

Fasting status, hydration, and acute stressors can temporarily alter lab values.